5 Modeling reality

This chapter covers

- Contextualizing data and linking it semantically to create a model of reality
- Ontologies as blueprints for semantic understanding of models of reality
- Knowledge graphs as a way to model relationships in the real world
- Serving the model of reality
- The importance of standardization

We have looked at how we can create a digital representation of a physical system, use sensors to measure how that system changes, and the ways in which we can integrate and store this data in a digital twin. But without some way to add meaning to this data, it remains a set of disconnected measurements that offer little insight into the true behavior of the physical system. Raw sensor readings such as temperature values, pressure measurements, or vibration frequencies are merely numbers until they are contextualized within an understanding of what they represent, how they relate to one another, and what patterns or anomalies might indicate about system health or performance.

The transformation from data to actionable intelligence requires layering understanding onto raw measurements. This involves creating relationships between diverse data streams and establishing the semantic frameworks necessary for a digital twin to interpret and reason about information, mirroring human expertise. This crucial step elevates the digital twin from a sophisticated monitoring tool to an intelligent system capable of reasoning and prediction.

We begin by examining how raw measurements become meaningful insights about system behavior. You'll discover how ontologies create semantic meaning, knowledge graphs build interconnected representations, and mathematical models provide quantitative foundations for analysis. By the end, you'll design modeling strategies that transform disparate data into